

Solution Requirements for Opti-Crop

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Requirement ID	Requirement	Description
FR-01	User Data Input	The system should allow users to enter crop, soil, weather, rainfall, temperature, humidity, and fertilizer details.
FR-02	Dataset Upload	The system should support loading agricultural datasets in CSV format for analysis.
FR-03	Data Validation	The system should validate user inputs and remove invalid or missing values before processing.
FR-04	Data Preprocessing	The system should clean, normalize, and prepare the dataset for machine learning.

Requirement ID	Requirement	Description
FR-05	Crop Recommendation	The system should recommend the most suitable crop based on environmental conditions.
FR-06	Yield Prediction	The system should predict expected crop yield using trained machine learning models.
FR-07	Pattern Identification	The system should identify trends and relationships between soil, weather, and crop production.
FR-08	Environmental Analysis	The system should analyze environmental factors affecting crop growth.
FR-09	Resource Optimization	The system should provide recommendations for efficient use of water, fertilizers, and nutrients.
FR-10	Decision Support	The system should generate recommendations to improve agricultural productivity.
FR-11	Data Visualization	The system should display graphs, charts, and analysis reports.
FR-12	Prediction Results	The system should display prediction accuracy, recommended crops, and yield estimates.
FR-13	Report Generation	The system should allow users to download prediction reports.

Requirement ID	Requirement	Description
FR-14	Model Management	The system should load and use trained machine learning models for predictions.
FR-15	Error Handling	The system should display meaningful error messages for invalid inputs or processing failures.

Software Requirements:

Component	Requirement
Programming Language	Python 3.x
Frontend	HTML, CSS, JavaScript
Backend	Flask
Database	SQLite
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Development Environment	VS Code / PyCharm